

A Novel Online Sketch Graphics Recognition Method for the Scene of Road Traffic Accident

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Abstract—A novel online sketch graphics recognition for the scene of road traffic accident is proposed to make the traffic accident processing more conveniently and efficiently. The practical human-machine interaction interface is designed to support showing and updating the corresponding feedback graphics on the screen as the operator drawing. The graphics recognition process is the crucial part of human-machine system. This paper proposes a novel way for sketch graphics recognition of traffic accident scene, the experimental results show the effectiveness of the proposed method.

Keywords—online sketchy graphics recognition; scene of road traffic accident; corresponding feedback; human-machine interaction

I. INTRODUCTION

As the essential evidence, the ichnography for the scene of road traffic accident is part and parcel in traffic accident processing. The normal drawing method includes two steps: (1) sketch rough drawing on coordinate paper at the accident scene, (2) protract national standards graphics with CAD software. The existing traffic accident graphics system with dragging, rotating and zooming function support operators to choose the standard graphics from the library[1,2]. But a large number of graphics elements restrain the operators from presenting their intention smoothly. To improve the efficiency, it is necessary to design and develop the easily operated online sketchy graphics recognition system for the traffic accident.

Sketch-based graphics input is regarded as the natural way to express the user's intention[3]. With the increasing availability of tablet notebooks and pen based PDAs, sketch-based interaction has become the indispensable part of perceptual user interface[4, 5]. Sketchy graphics recognition requires excavating effective data from irregular freehand sketching and includes two parts: stroke preprocessing and shapes recognition. Stroke preprocessing is a common procedure to optimize original input, then shapes recognition place processed input in different categories. Sketch recognition is the crucial part in traffic accident human-machine interaction system. According to various graphics processing mechanisms, the existing graphics recognition can be divided into four types : strokes-based method, primitive shapes-based method, geometric features-based method and composite shapes-based method [6] which are explained as following:

- Strokes-based method. Early recognition system treated strokes equivalent as graphics, and the classical Rubine model is applied in many systems[7]. But the system is restricted to single-strokes recognition.
- Primitive shapes-based method. By this method, the complex intricate graphics is segmented into the combination of several primitive shapes, such as straight line, polyline, circle and arc. Each kind of shape can be distinguished through curvature and velocity[8].After being fitted and regularized, the combined shapes can be recognized by the matching template. The correctness of segmentation directly affects the efficiency of the system.
- Geometric features-based method. The sketch graphics is treated as a whole entity, and is normalized into the $n*n$ meshes, the geometric features are got from each mesh [6, 9]. The essence of geometric feature extraction is to reduce the dimension, therefore part of feature may lost and the recognition accuracy is low.
- Composite shapes-based method. The recognition process introduces several simple and common geometric figures as reference to improve the accuracy [9]. The graphics is firstly mapped to different attributes, then the system predicts the following valid status with other constraints. This method meets the requirement of complicated graphics recognition with high matching rate.

In this paper, we proposed an effective method for the traffic graphics sketch recognition. We proposed the graphic recognition method for the five kinds of basic graphics: circle, triangle, quadrilateral, parallel lines and irregular figure and one closed graphics search algorithm. All the graphics can be classified into five categories by approximating its overall configuration as five basic figures. Firstly, the rough recognition is realized by detecting five basic figures from input data, meanwhile the similarity-based corresponding feedback suggest the user's intention. Furthermore the other data can assist to subdivide to improve feedback accuracy. Such a system would make the traffic accident processing more quickly and conveniently.

This paper is organized as following. Section II gives the classification of basic graphics and strokes. Section III introduces the graphics recognition method for the five kinds of

basic graphics and the closed graphics search algorithms. Section IV introduces the whole Sketch Graphics Recognition System. Section V shows the experiment of the system. Finally, the conclusion is given in section VI.

II. GRAPHICS AND STROKES CLASSIFICATION

There is no defined set of shapes to be recognized in online sketch graphics recognition system, but the ordinary mode of sketch shapes can be divided into two categories: open strokes and closed strokes[10]. Open strokes can be segmented into lines and arcs. Closed strokes includes circle, triangle and quadrilateral drawn in single- strokes.

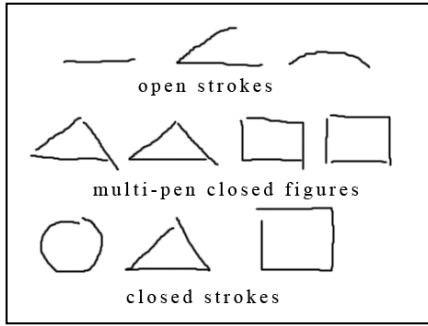


Figure 1. The basic elements of the sketch graphics.

According to the features of traffic graphics library, the graphic can be approximated as one of the five basic figures, as is illustrated in figure 2. What we analyzed is the incomplete sketches. To realize graphics similarity-based corresponding feedback recognition, five basic figures should be detected from user's input shapes firstly.

TABLE I. GRAPHIC SYMBOLS OF THE SCENE OF TRAFFIC ACCIDENT

Circle	Triangle	Quadrilateral	Parallel lines	Irregular figure

III. GRAPHIC RECOGNITION METHOD AND CLOSED GRAPHICS SEARCH ALGORITHMS

Human-machine interaction interface shows the track of freehand drawing on the screen while tracing pen's movement. And the system record original data: the coordinate and the time of the sample. A single stroke is expressed as a set of sample point $(N_1, N_1 \dots N_n)$, between the interval of mouse-down and the mouse-up. The input sample data contains info of

strokes' distinction and users' input habit, after being preprocessed, features for subsequent recognition can be collected by recognition algorithms.

A. Open strokes Recognition

1) *Straight Line Recognition*: Mathematically, straight line is the shortest distance between start and end point. The recognition based on the ratio of:

$$\gamma = \frac{Dis(N_1, N_n)}{\sum_{i=1}^{n-1} Dis(N_i, N_{i+1})} \quad (1)$$

where N_i denotes the i th sample point, $Dis(N_1, N_n)$ is the distance between the first and last sampling point, $\sum_{i=1}^{n-1} Dis(N_i, N_{i+1})$ is the graph's length. If the ratio γ is less than the threshold γ_1 , the graph can be recognized as straight line. And the slopes:

$$k = (N_{ny} - N_{1y}) / (N_{nx} - N_{1x}) \quad (2)$$

can be as the judgment criterion for the parallel lines.

2) *Polygonal Line Recognition*: Polygonal line can be considered as gather of the straight lines linked by vertices, thus the crucial step of polygonal line recognition is to detect the vertices.

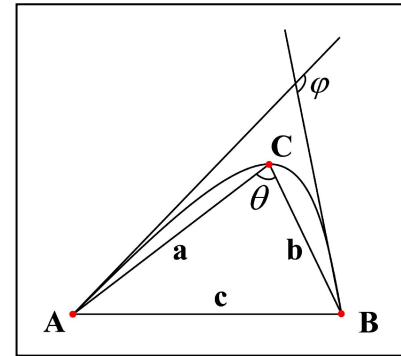


Figure 2. Curvature of polygonal line.

Curvature is the change rate of the angle between a curve and a tangent to the curve, which can defined as:

$$k = \frac{\Delta\phi}{\Delta s} \quad (3)$$

When sampling interval is short enough, $a \rightarrow 0, b \rightarrow 0$, $\Delta\phi \rightarrow \pi - \theta$,

$$k = \frac{\Delta\varphi}{\Delta s} \approx \frac{\pi - \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right)}{a + b} \quad (4)$$

Intuitively, vertices are marked only at the corners of the stroke, so vertices are the endpoints of linear segments of the stroke where the local curvature is high[8]. If the point's k is above the threshold k_l , the local point can be recognized as vertex.

3) *Arc Recognition*: Arc is a continuous portion of a circle, the feature is somewhere between straight line and polygonal line. Therefore arc recognition can base on the recognition method of above two figures. If the ratio γ is above the threshold γ_l and all the sampling points' curve are below the threshold k_l , the open stroke can be recognized as arc.

B. Closed strokes Recognition

1) *Circle Recognition*: Sketch circle path conform circle's mathematical definition that a plane curve generated by one point moving at a constant distance from a fixed point. Sketch's centroid[11] is expressed as:

$$P_N = \frac{1}{n} \sum_{i=1}^n N_i \quad (5)$$

sketch's centroid radii is expressed as:

$$R_N = \frac{1}{n} \sum_{i=1}^n |N_i - P_N| \quad (6)$$

mean-square error of centroid radii is expressed as:

$$\sigma_N = \sqrt{\frac{1}{n} \sum_{i=1}^n (R_n - (N_i - P_N))^2}. \quad (7)$$

For sketch circle, the centroid radii should be steady than others, If σ_N is less than the threshold σ_{N1} , the graph can be recognized as circle.

2) *Triangle and Quadrilateral Recognition*: Mean-square error of centroid radii σ_N can distinguish the sketch circle and open strokes with high recognition rates, but failed to differentiate circle, triangle, and quadrilateral. As what we analyzed is the sketches with noise which is drawn hastily in limited time, optimal threshold σ_{N1} for circle, triangle and quadrilateral is hard to derive. To solve these problems, we use both the combine curvature and centroid radii mean-square error characteristic as the recognition criteria. Closed sketch graphics with high curvature point and low centroid radii mean-square error can be recognized as triangle or quadrilateral. Moreover the number of vertices can be used to differentiate triangle and quadrilateral.

C. Closed Graphics Search Algorithms

Closed graphics search algorithms is the critical part of the

sketch recognizers, it is introduced to search the basic figures composed by several open strokes. The key is to obtain the open strokes optimal combination. Given definition as:

$$D_{s-s} = Dis(N_{i-s}, N_{j-s}) \quad (8)$$

the distance between stroke i start point and stroke j start point;

$$D_{s-e} = Dis(N_{i-s}, N_{j-e}) \quad (9)$$

the distance between stroke i start point and stroke j end point.

$$D_{e-e} = Dis(N_{i-e}, N_{j-e}) \quad (10)$$

the distance between end i start point and stroke j end point.

Optimal search algorithms is as follows:

- choose m open strokes from input graphics;
- if D_{s-s} , D_{s-e} , D_{e-e} is less than the threshold D_l , $tem = tem + 1$;
- if $tem = 2 \times m$, search finished, or else start from the first step.

The vertices number of searched closed graphics is decided by stroke number m and the vertices number of every open stroke.

IV. THE SCHEME OF SKETCH GRAPHICS RECOGNITION SYSTEM

A scheme of sketch interface which can show and update the corresponding graphics in real time is described in Fig. 3. The sketch recognition process can be divided into the following four steps:

- **Strokes Classification**: users' sketch shapes can be divided into two categories: open strokes and closed strokes, and open strokes are recognized as lines and arcs.
- **Shape recognition**: closed strokes are recognized as circle, triangle, quadrilateral by judgment basis of curvature and centroid radii mean-square error. And the five basic figures composed by open strokes are obtained by closed graphics search algorithms.
- **Similarity-based corresponding feedback**: the rough recognition is realized by detecting five basic figures, the other open strokes can assist to subdivide to improve feedback accuracy.
- **Human-machine interaction**: user can perform operations of confirm, delete, redo, undo.

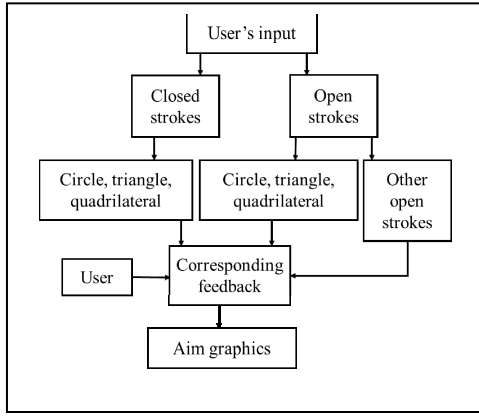


Figure 3. Scheme of sketch graphics recognition system.

V. EXPERIMENT AND EVALUATION

The experiment selects 97 general standard traffic graphics according to GB/T11797-2005 "The graphics symbols for the scene of road traffic accident".

In this experiment, the parameters are chosen as follows: interval of sketch point series is 5 pixels, $\gamma_1 = 0.9$, $k_1 = 0.7$, $\sigma_{N1} = 9.5$. Three random persons are asked to draw graphics, and each category of graphics are drawn for 50 times. The obtained recognition ratio is shown in Table II. The graphics recognition process is shown in Table III.

TABLE II. RECOGNITION RATIO FOR DIFFERENT USERS

User	Categories of Traffic Accident Sketch Graphics				
	<i>Circle</i>	<i>Triangle</i>	<i>Quadrilateral</i>	<i>Parallel lines</i>	<i>Irregular figure</i>
1	98%	92%	88%	90%	87%
2	94%	90%	84.5%	87%	84%
3	92%	86%	84%	85%	83%

TABLE III. CORRESPONDING FEEDBACK

Steps	Input	Corresponding Feedback			
1					
2					
3					

VI. CONCLUSION

In this paper, we proposed a system of online sketch

graphics recognition method for traffic accident scene, which is capable of using composite shapes-based method by classifying the graphics into five categories according to its overall configuration. The experimental results show the system can successfully realize the sketch graphics recognition which allows users to draw in an unrestricted fashion, moreover the similarity-based corresponding feedback enabling a more natural interaction with the computer. Such a system would make the traffic accident processing more quickly and conveniently.

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